El-Shorouk Academy **Higher Institute of Engineering**		**Department:** **Department of Communication and Computer Engineering**
Course Name: selective course (6) (Network security and cryptography)	هندسة الاتصالات والحاسبات Communications and Computers Engineering Dept.	**Level: 4th Year**
Course Code: CCE		**Semester: Second 2023/2024**
Course Teacher: Dr/ Bassam Wasfi		**Sheet No:2**

Solve the following questions

A) Solve the following questions :

1. Suppose that your friend suggests the following way to confirm that both of you have the same secret key. You create a random bit string which is having the length of the key, then XOR it with the key, and send the result over the channel. Your partner XORs the incoming block with the key (which should be the same key) and sends the result back. If what you receive is your original string, you have verified that your partner has the same secret key, yet neither of you has ever transmitted the key on the channel. Is there a flaw in this scheme?

Yes. An attacker eavesdropping on the traffic will get two strings (Key ⊕ Random) and (Random), one sent in each direction, and from the properties of XOR, the attacker can just XOR the two values and get the secret key.

2. Fill in the table below, according to the following assumptions: Modern computers run approximately 3 billion compute cycles per second, and with cloud computing, it is easily imaginable that a well-funded private organization can rent 100,000 such systems in the cloud. To test whether a given ciphertext decrypted with a key matches a known plaintext takes approximately 1000 compute cycles.

a. Assuming that we have a block of ciphertext with known plaintext, indicate the time in years needed to test every possible key for each key length, in the case of:
   I. One modern computer.
   II. A well-funded private organization as described above.

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Key Length (bits)	Number of Keys	Attack Time (Years) (Single computer)	Attack Time (Years) (private organization (100000 computers))
56			
128			
256			
512			

Key Length (bits)	Number of Keys	Attack Time (Years) (Single Computer)	Attack Time (Years) (Private Organization (100,000 Computers))
n	2^n	$2^n/(3*10^6*31536000)$	$2^n/(3*10^{11}*31536000)$
56	7.205759404×10^{16}	761.643772598	0.007616438
128	3.402823669×10^{38}	3.596761024×10^{24}	3.596761024×10^{19}
256	1.157920892×10^{77}	1.223914354×10^{63}	1.223914354×10^{58}
512	$1.340780793\times10^{154}$	$1.417196001\times10^{140}$	$1.417196001\times10^{135}$

b. Assume you were working on a research lab, and you were asked to find a solution to improve the cracking performance and time. Your team came up with two promising solutions.

I. Use specialized hardware to increase the speed of such an attack by 1000 times.

II. Make cryptanalysis of the cipher to be able to guess half of the key bits at little cost. Explain with calculations which solution do you think would give better results

Key Length (bits)	Number of Keys	Attack Time (Years) (increase 1000 times)	Attack Time (Years) (cryptanalysis half key)
56			
128			
256			
512			

Assume we will use the single computer case to test our analysis for the two research cases:

Key Length (bits)	Attack Time (Years) (increase 1000 times)	Attack Time (Years) (cryptanalysis half key)
n	$2^n/(3*10^9*31536000)$	$2^{n/2}/(3*10^6*31536000)$
56	0.761643773	0.000002837
128	3.596761024×10^{21}	194980.805785024
256	1.223914354×10^{60}	3.596761024×10^{24}
512	$1.417196001\times10^{137}$	1.223914354×10^{63}

From our analysis we find that using cryptanalysis of the cipher to guess half of the key bits at little cost, gives way better results than increasing the speed of hardware by 1000 times.

3. Use the Caesar cipher to encrypt and decrypt the message "Network security and cryptography course" and the key (shift) value of this message is 5,then decrypt it to approve your answer.

P: Plain text
K: key
C: cipher text

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(3) P: Network security and cryptography course

K: 5 Caesar C: (K+P) mod 26 =

P: a, C: (5+0) mod 26 = 5 → f

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e

decreption

encryption التشفير

(P) Plain : حروف المقابل

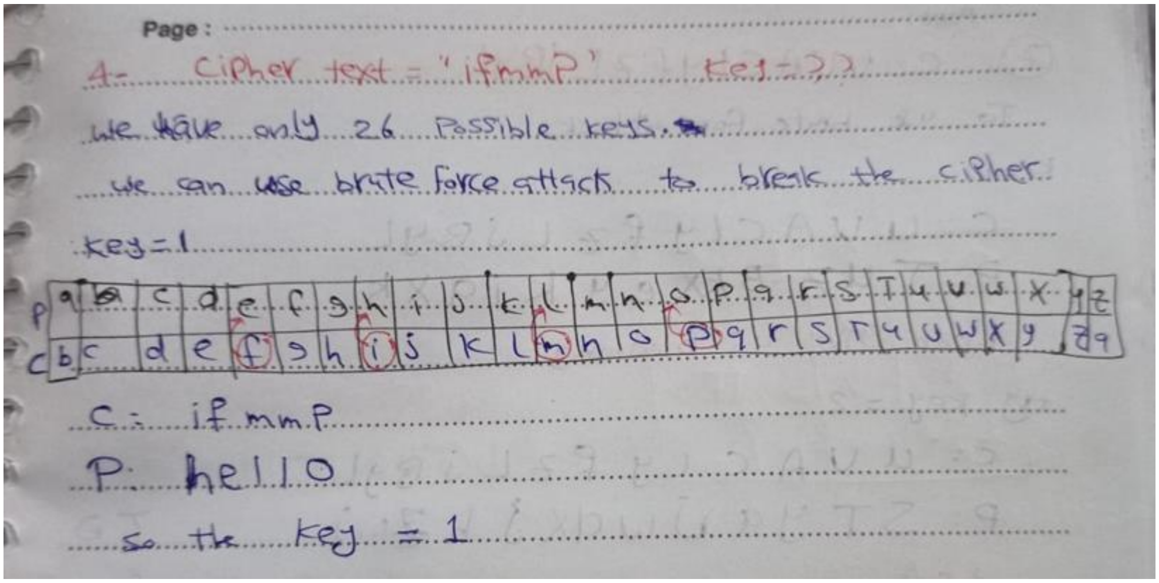
P: Network Security and cryptography Course

C: Sjyb Twp xjhawnyd fsi hwduyTLWfumd hTZWxj

N ← S : decreption لو عايز

(P) Plain ≠ (C) cipher المقابل العدد حروف

4. Use the Caesar cipher to find the plaintext and the key from the cipher text:
Cipher text : "ifmmp"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4- cipher text = 'ifmmP' key = ??

we have only 26 possible keys.

we can use brute force attack to break the cipher.

key = 1

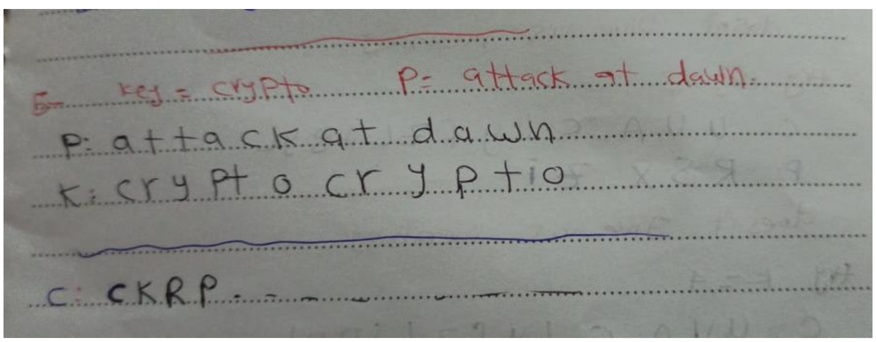
P	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
C	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a

C: ifmmP

P: hello

so the key = 1

5. Use the Vigenere cipher with key = crypto to encrypt the message “attack at dawn”

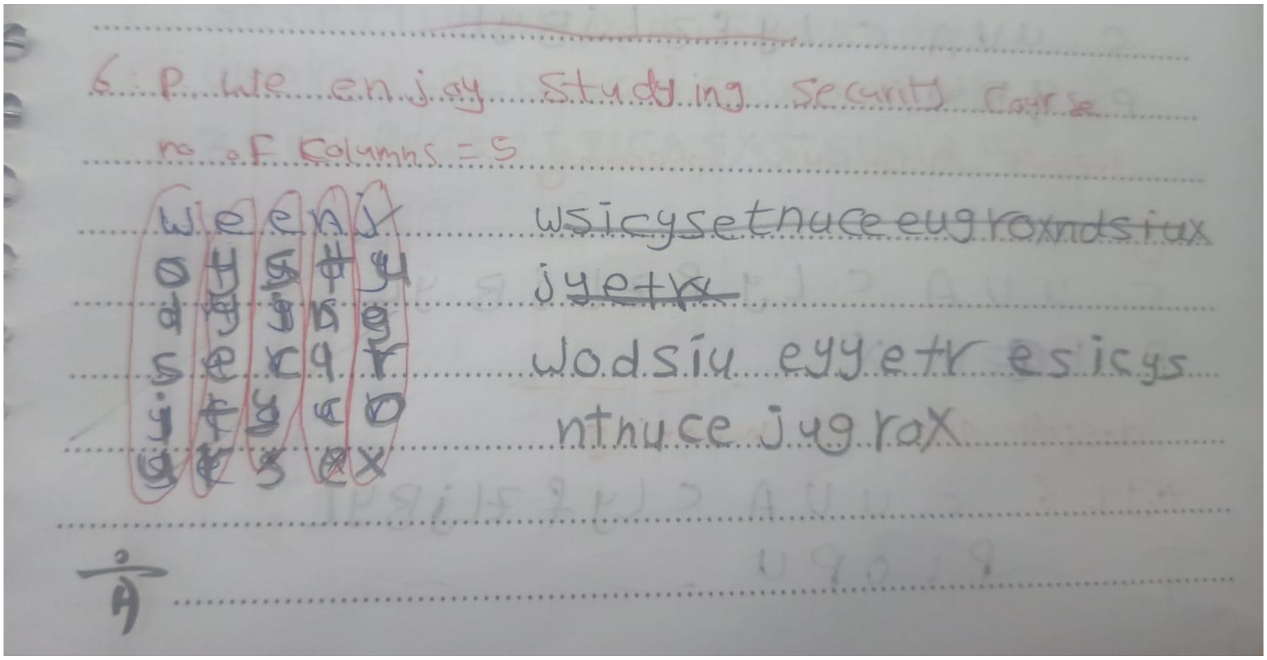

5- key = crypto P: attack at dawn.

p: attack at dawn

k: crypto crypto

C: CKRP...

6. Encrypt the following plaintext using Columnar transposition: We enjoy studying security course. Columns is 5



7. Your friend has intercepted the cipher text "UVACLYFZLJBYL". Show how he/she can use a brute- force attack to break the cipher.

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(7) C: UUACLyfzLjByL

To use brute-force attack

K=1

C: UVACLy fzLjByL

P: Tuz BkXey*Kiaxk

doesn't give us meaning

try key = 2

C: UVACLy PzLjByL

P: STyaJwdxJhzij

doesn't give us meaning

try key = 3

C: UUA CLy fzLjByL

P: RSX zi _ _ _

doesn't give us meaning

try K = 4

C: UVA CLyfzLJByL

P: qRW _ _ _ _ _

doesn't give us meaning

try K = 5

C: UUA CLyPzLjByL

P: P[illegible]v X _ _

doesn't give us meaning

try K = 6 C: UUA CLyfzLjByL

P: OPU _ _ _

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doesn't give us meaning

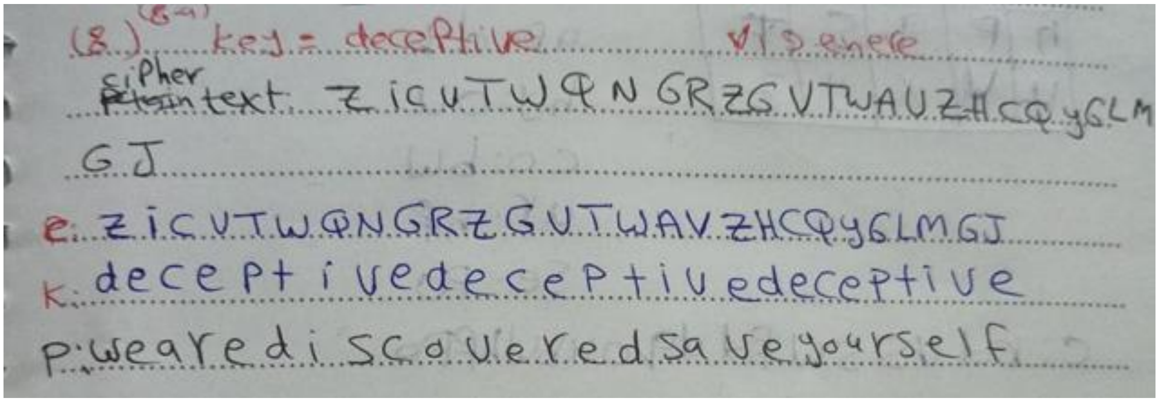
try K = 7

C: UVACLyP zL JByL

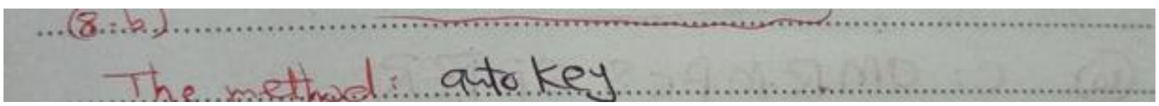
P: noTVery secure

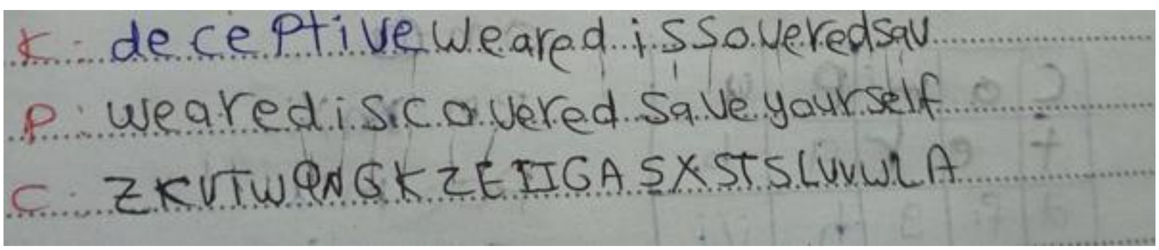
Key = 7

8. A. Use the Vigenere cipher with key = deceptive to decrypt the message "ZICVTWQNGRZGVTWAVZHCQYGLMGJ" .


(8.) (8-a) key = deceptive vigenere
cipher text ZICVTWQNGRZGVTWAVZHCQYGLM
GJ
e: ZICVTWQNGRZGVTWAVZHCQYGLMGJ
k: deceptivedeceptivedeceptive
p: wearediscoveredsaveyourself

B. In this method letter frequencies are obscured so vigenere proposed another method what is this method and use it to encrypt the text you have been decrypted from a.


(8.b)
The method: auto key


k: deceptiveWearedissoveredsav
p: wearediscoveredsaveyourself
c: ZKVTWQNGKZEIIGASXSTSLUVWLA

9. Use the playfair cipher method with key =hello world to encrypt the message "cryptography course"

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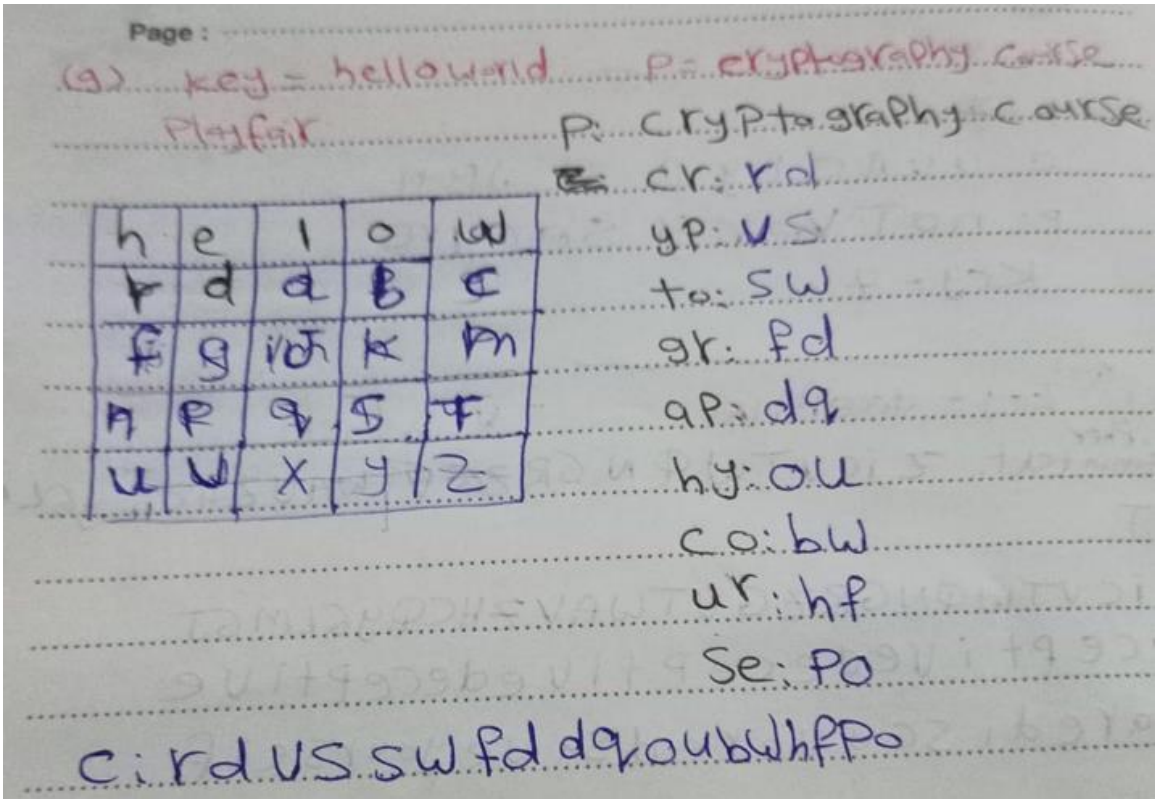

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(9) key = helloworld P = cryptography course

Playfair

P: cryptography course

h	e	l	o	w
r	d	a	b	c
f	g	i/j	k	m
n	p	q	s	t
u	v	x	y	z

cr: rd

yp: vs

to: sw

gr: fd

ap: dq

hy: ou

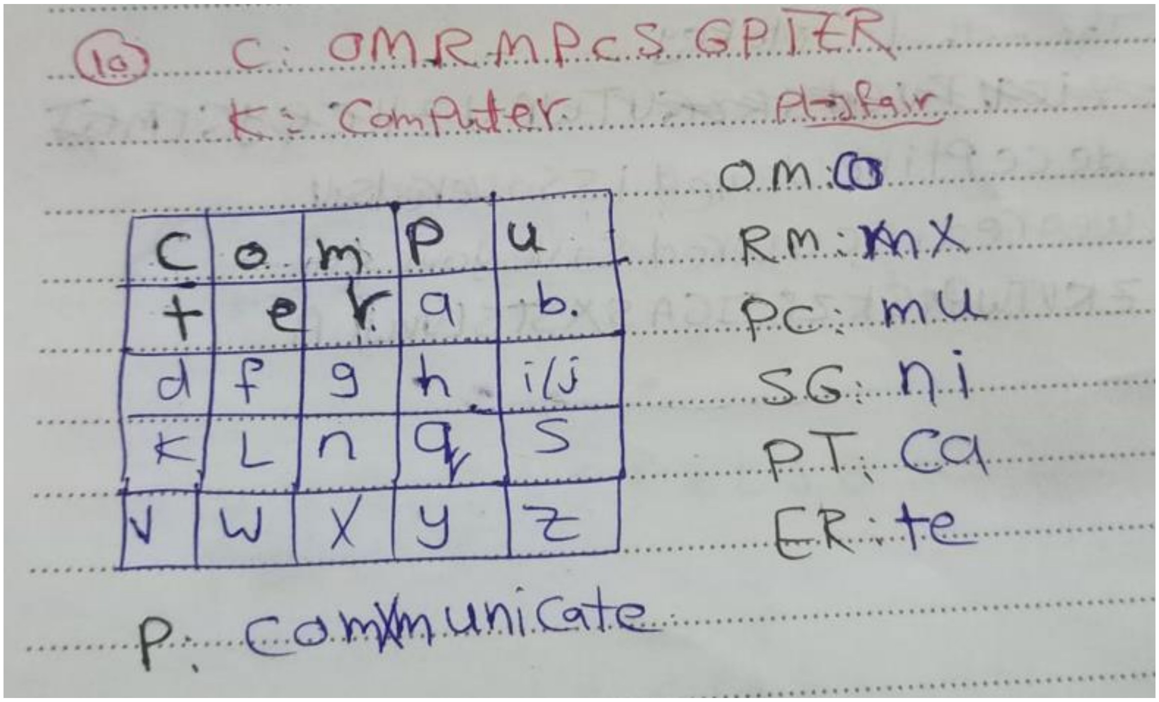
co: bw

ur: hf

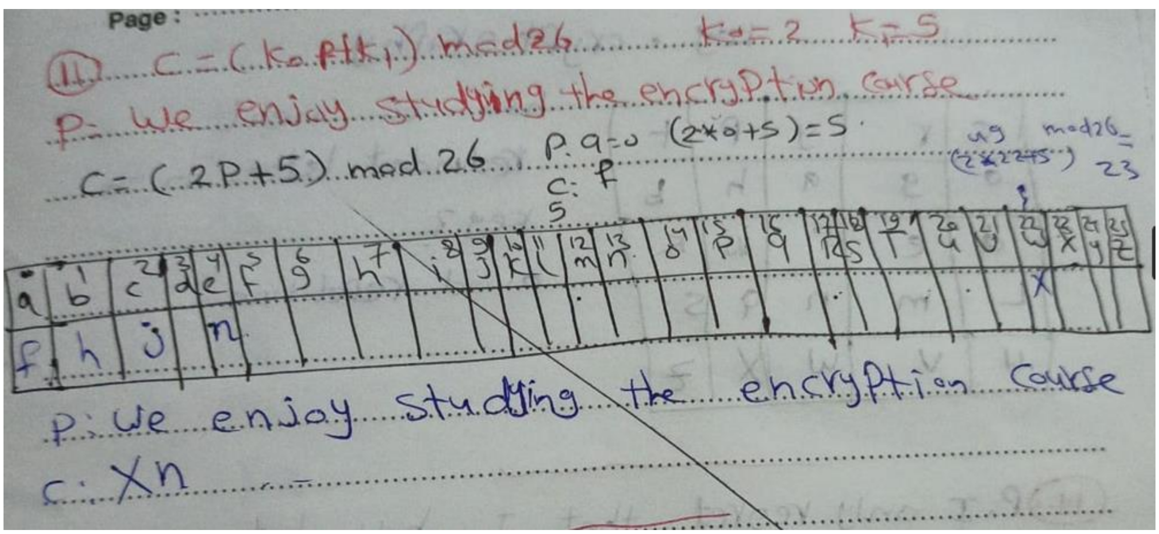
se: po

c: rdvssw fd dqoubwhfpo

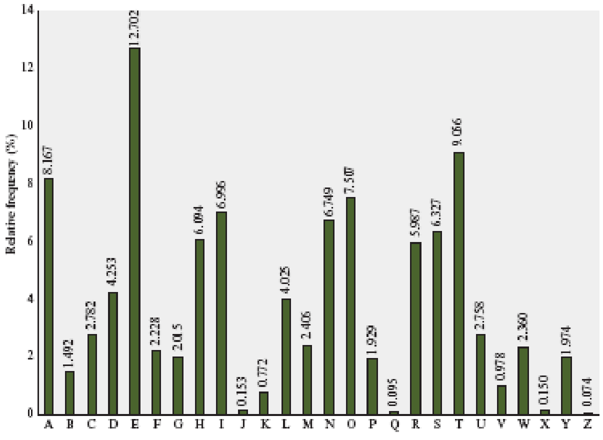
10. Assume "OMRMPCSGPTER" is the ciphertext and “computer” is the encryption key. Using the playfair cipher method to find the plain text.

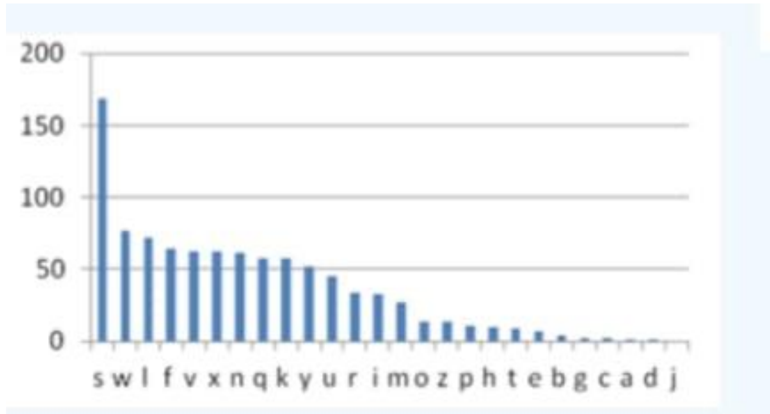


11. Encrypt the following text "we enjoy studying the encryption course" where C=(k0 p+ k1) mod 26. Such that: k0 = 2, k1 = 5.

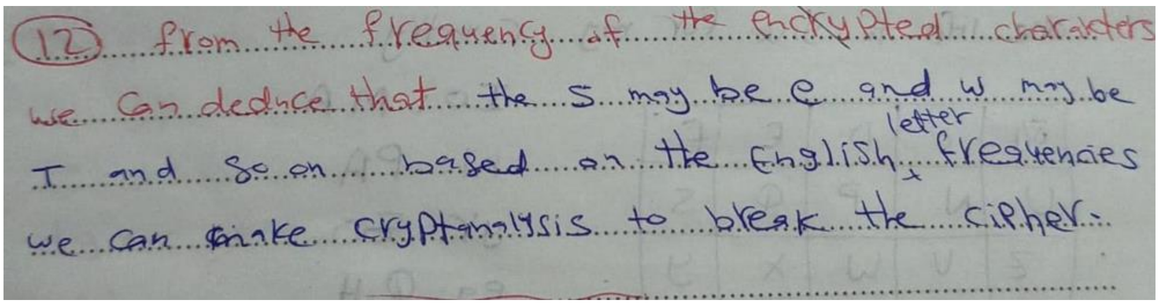


12. Suppose that you have the frequency of different characters in some encrypted text such as this chart in figure 1 What can you deduce about the mapping?



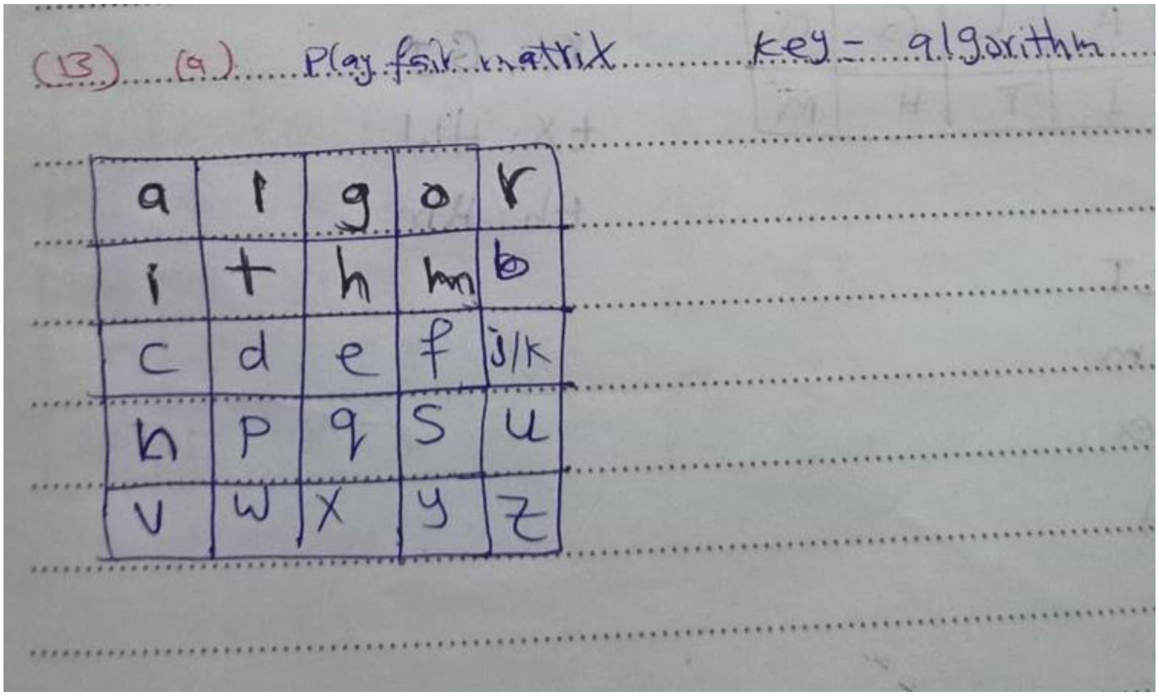


English Letter Frequencies Figure 1


(12) from the frequency of the encrypted characters we can deduce that the s may be e and w may be T and so on based on the English letter frequencies we can make cryptanalysis to break the cipher.

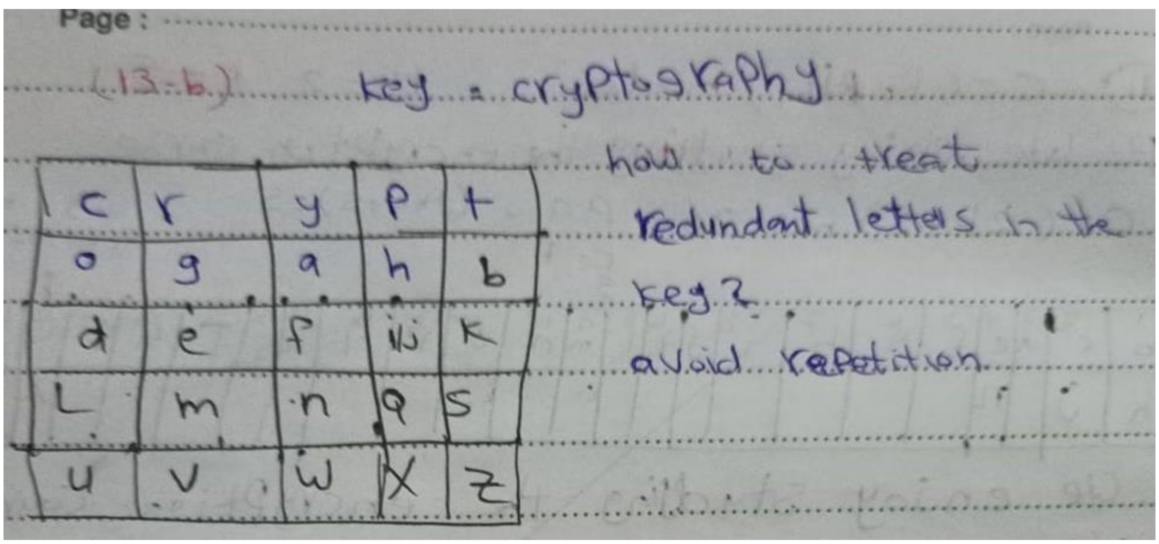
13.

a. Construct a Playfair matrix with the key *algorithm*.
b. Construct a Playfair matrix with the key *cryptography*. Make a reasonable assumptionabout how to treat redundant letters in the key.



(13) (a) Play fair matrix key = algorithm

a	l	g	o	r
i	t	h	m	b
c	d	e	f	j/k
n	p	q	s	u
v	w	x	y	z



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(13-b) key = cryptography

how to treat redundant letters in the key?

avoid repetition

c	r	y	p	t
o	g	a	h	b
d	e	f	i/j	k
l	m	n	q	s
u	v	w	x	z

14.

a. Using this Playfair matrix to encrypt this message : I only regret that I have but one life to give for my country.

J/K	C	D	E	F
U	N	P	Q	S
Z	V	W	X	Y
R	A	L	G	O
B	I	T	H	M

b. Repeat part (a) using the Playfair matrix from Problem 13.a.

14.b P: I only regret that I have but one life to give for my country

J/K	C	D	E	F
U	N	P	Q	S
Z	V	W	X	Y
R	A	L	G	O
B	I	T	H	M

Io: MA
nl: PA
yr: ZO
eg: QH
re: GJ
tx: HW
th: HM
at: LI
Ih: TM
av: IA
eb: JH
ut: Pb
on: aS
el: DG

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Page :

if : MC

et : DH

og : RO

iv : CA

ef : fJ

or : RA

my : fo

co : fa

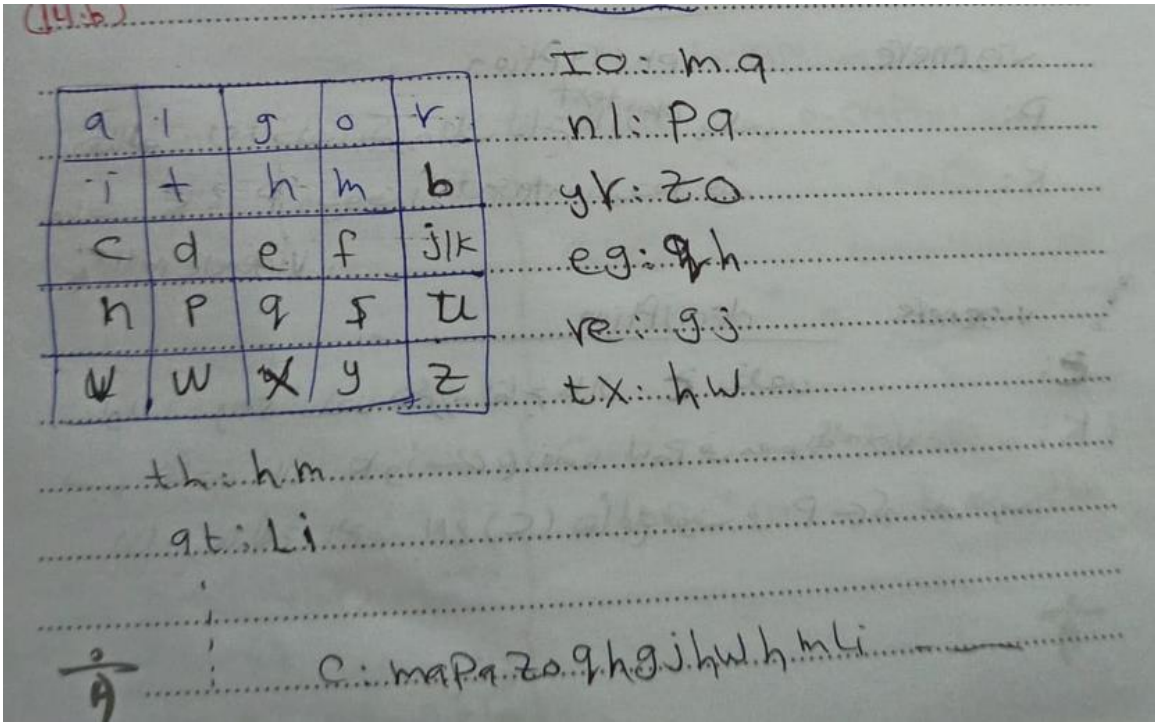
un : NP

tr : bL

yx : Zy

C : MAPa Z oE H GJ HWH mLI Tm ta JH Pb aS

dg mc DH Ro CA FJ RA fo fa NP bL Zy



15.

A . Given the keyword SECURITY, encrypt the following plaintext using the autokey system. : “We are discovered. Save yourself”

B. Given a long ciphertext which has been encrypted with the autokey system, how can one find the secret key used?

16. Using the Vigenère cipher, encrypt the word “cryptographic” using the word “eng”.

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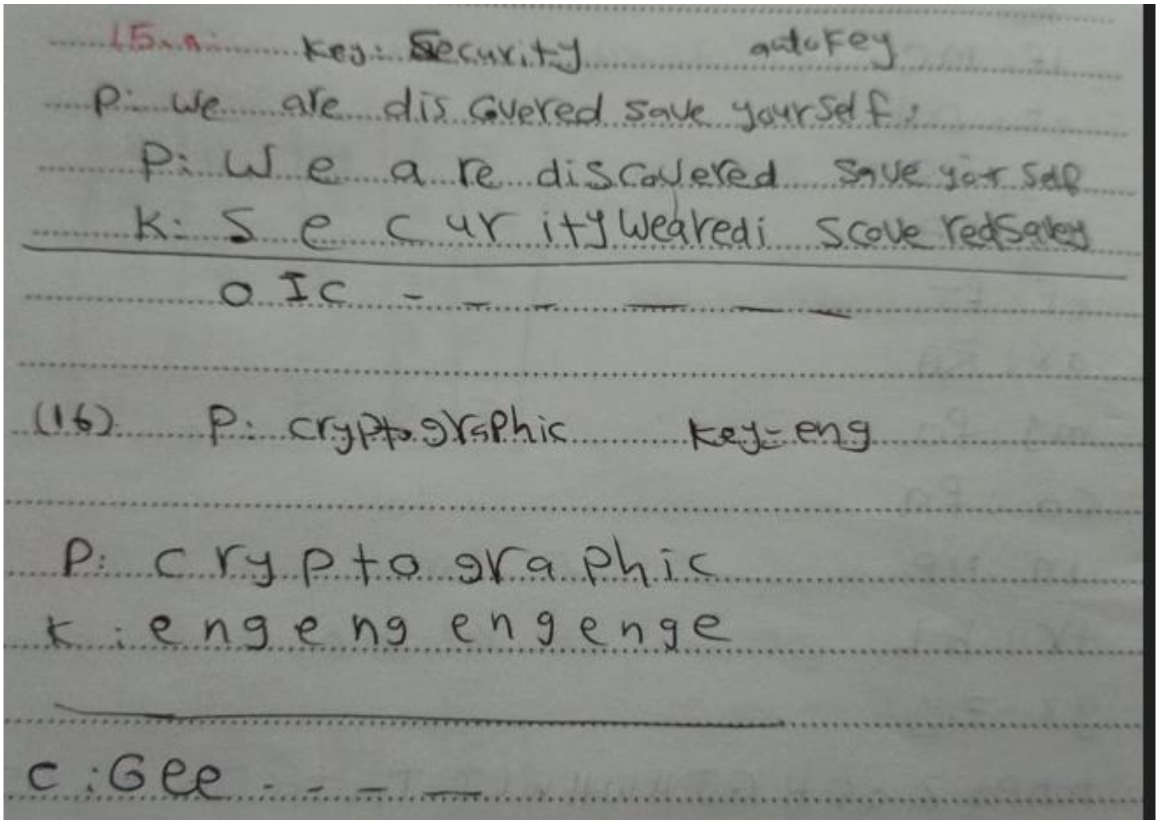
15.a: Key: Security autokey
P: we are dis covered save yourself:
P: w e a re discovered save yor self
K: S e c ur ity weared i scove redsavey
C: I C - - - - -

(16) P: cryptographic key= eng
P: c r y p t o g r a p h i c
K: e n g e n g e n g e n g e
C: Gee - - - -

B)MULTIPLE CHOICE QUESTIONS:

1. _________ is the science and art of transforming messages to make them secure and immune to attacks.
   A. Cryptography
   B. Cryptoanalysis
   C. either (a) or (b)
   D. neither (a) nor (b)

2. The _________is the original message before transformation

   A. ciphertext
   B. plaintext
   C. secret-text
   D. none of the above

3. The _________ is the message after transformation

A. ciphertext
B. plaintext
C. secret-text
D. none of the above

4. A(n) _______ algorithm transforms plaintext to ciphertext

A. encryption

B. decryption

C. either (a) or (b)

D. neither (a) nor (b)

5. The _______ is a number or a set of numbers on which the cipher operates

A. Secret
B. cipher
C. key
D. none of the above

6. *A* ________ cipher replaces one character with another character.

A. Transposition
B. substitution
C. either (a) or (b)
D. neither (a) nor (b)

7. __________ ciphers can be categorized into two broad categories: monoalphabetic and polyalphabetic.

   A. Transposition
   B. Substitution
   C. either (a) or (b)
   D. neither (a) nor (b)

8. The ________ cipher is the simplest monoalphabetic cipher. It uses modular arithmetic with a modulus of 26.

   A. Additive
   B. transposition
   C. shift
   D. none of the above

9. The Caesar cipher is a ________cipher that has a key of 3.

   A. Additive
   B. transposition
   C. shift
   D. none of the above

10. The _________ cipher reorders the plaintext characters to create a ciphertext.

   A. substitution
   B. transposition
   C. either (a) or (b)
   D. neither (a) nor (b)

11. Shift cipher is also referred to as the

   A. Caesar cipher

B. cipher text
C. Shift cipher
D. None of the above

12. Use Caesar's Cipher to decipher the following
HQFUBSWHG WHAW
a)ABANDONED LOCK
b) ENCRYPTED TEXT
c) ABANDONED TEXT
d) ENCRYPTED LOCK

13. Caesar Cipher is an example of
a) Poly-alphabetic Cipher
b) Mono-alphabetic Cipher
c) Multi-alphabetic Cipher
d) Bi-alphabetic Cipher

14. On Encrypting "thepepsiisintherefrigerator" using Vignere Cipher System using the keyword "HUMOR" we get cipher text-
a) abqdnwewuwjphfvrrtrfznsdokvl
b) abqdvmwuwjphfvvyyrfznydokvl
c) tbqyrvmwuwjphfvvyyrfznydokvl
d) baiuvmwuwjphfoeiyrfznydokvl

15. On Encrypting "cryptography" using Vignere Cipher System using the keyword "LUCKY" we get cipher text
a) nlazeiibljji
b) nlazeiibljii
c) olaaeiibljki
d) mlaaeiibljki

16. What is the meaning of cipher in computer terminology?
a) an algorithm that performs encryption
b) an algorithm that generates a secret code
c) an algorithm that performs encryption or decryption
d) a secret code

17. Which of the following cipher is created by shuffling the letters of a word?
a) substitution cipher

b) transposition cipher
c) mono alphabetic cipher
d) poly alphabetic cipher

18. In which of the following cipher the plain text and the ciphered text have same set of letters?
a) one time pad cipher
b) columnar transposition cipher
c) playfair cipher
d) All the above

19. In which of the following cipher the plain text and the ciphered text does not have same number of letters?
a) caesar cipher
b) playfair cipher
c) transposition cipher
d) Monoalphabetic cipher

20. Which of the following ciphers are created by shuffling the letters of a word?
a) substitution cipher
b) transposition cipher
c) vigenere cipher
d) one time pad cipher

21. Which of the following correctly defines poly alphabetic cipher?
a) a substitution based cipher which uses multiple substitution at different positions
b) a substitution based cipher which uses fixed substitution over entire message
c) a transposition based cipher which uses multiple substitution at different positions
d) a transposition based cipher which uses fixed substitution over entire message